

[Paper 12 continued from p. 78. Page 79 of the printed volume.]

I proceed to prove a theorem, which may be expressed as follows:

$$[A.k.2p]^+ \cdot [B.k.2q]^+ = [AB | k.2p + 2q - 2]^+, \quad (27)$$

where

$$\overline{AB} |_{rs\dots xy\dots} = S \cdot A_{rs\dots l} B_{xy\dots l}, \quad (28)$$

the number of the symbols $r, s, \dots$ being obviously $2p-1$, and that of $x, y, \dots$ being $2q-1$. The summatory sign S refers to l , and denotes the sum of the several terms corresponding to values of l from $l = 1$ to $l = k$. Also the theorem would be equally true if l had been placed in any position whatever in the series $r, s \dots l$; and again, in any position whatever in the series $x, y \dots l$, instead of at the end of each of these. With a very slight modification this may be made to suit the case of an odd number instead of one of the two even numbers $2p, 2q$; (in fact, it is only necessary to place the mark $(+)$ in $\{AB | \dots\}$ over the column corresponding to the marked column in $\{A\dots\}, \{B\dots\}$ being the symbol for which the number of columns is odd), but it is inapplicable where the two numbers are odd. Consider the second side of (27); this may be expanded in the form

$$\Sigma \pm_s \dots \pm_x \pm_y \dots \overline{AB} |_{s_1\dots x_1 y_1\dots} \overline{AB} |_{s_2\dots x_2 y_2\dots} \dots \overline{AB} |_{s_k\dots x_k y_k\dots}, \quad (29)$$

where Σ refers to the different quantities $s, \dots, x, y, \dots$ as in (11).

Substituting from (28), this becomes

$$\Sigma \cdot S_{l_1} \dots S_{l_k} (\pm_s \pm_x \pm_y \dots A_{rs_1\dots l_1} \dots A_{rs_k\dots l_k} B_{xy_1\dots l_1} \dots B_{x_k y_k \dots l_k}). \quad (30)$$

Effecting the summation with respect to $x, y \dots$ this becomes

$$\Sigma \cdot S_{l_1} \dots S_{l_k} \pm_s \dots A_{rs_1\dots l_1} \dots A_{rs_k\dots l_k} \left\{ \begin{array}{c} \overline{B} 1 1 \dots l_1 \\ \vdots \\ k k \dots l_k \end{array} \right\}, \quad (31)$$

S now referring to $s, \dots$ only. The quantity under the sign S vanishes if any two of the quantities l are equal, and in the contrary case, we have

$$\left\{ \begin{array}{c} \overline{B} 1 1 \dots l_1 \\ \vdots \\ k k \dots l_k \end{array} \right\} = \pm_l [\overline{B}.k.2q]^+, \quad (32)$$

which reduces the above to

$$[\overline{B}.k.2q]^+ \cdot \Sigma \pm_s \dots \pm_l A_{rs_1\dots l_1} \dots A_{rs_k\dots l_k}, \quad (33)$$

Σ referring to the quantities $s \dots$, and also to the quantities l . And this is evidently equivalent to

$$[A.k.2p]^+ [\overline{B}.k.2q]^+, \quad (34)$$

the theorem to be proved. It is obvious that when $p = 1, q = 1$, the equation (27) coincides with the theorem (Θ), quoted in the introduction to this paper.

13.

ON THE THEORY OF LINEAR TRANSFORMATIONS.

[From the Cambridge Mathematical Journal, vol. iv. (1845), pp. 193–209.]

The following investigations were suggested to me by a very elegant paper on the same subject, published in the *Journal* by Mr Boole. The following remarkable theorem is there arrived at. If a rational homogeneous function U , of the n^{th} order, with the m variables $x, y, \dots$, be transformed by linear substitutions into a function V of the new variables $\xi, \eta, \dots$; if, moreover, θU expresses the function of the coefficients of U , which, equated to zero, is the result of the elimination of the variables from the series of equations $d_x U = 0$, $d_y U = 0$, &c., and of course θV the analogous function of the coefficients of V : then $\theta V = E^a \cdot \theta U$, where E is the determinant formed by the coefficients of the equations which connect $x, y, \dots$ with $\xi, \eta, \dots$, and $a = (n-1)^{m-1}$.¹ In attempting to demonstrate this very beautiful property, it occurred to me that it might be generalised by considering for the function U , not a homogeneous function of the n^{th} order between m variables, but one of the same order, containing n sets of m variables, and the variables of each set entering linearly. The form which Mr Boole's theorem thus assumes is $\theta V = E_1^{a_1} \cdot E_2^{a_2} \dots E_n^{a_n} \cdot \theta U$. This it was easy to demonstrate would be true, if θU satisfied a certain system of partial differential equations. I imagined at first that these would determine the function θU , (supposed, in analogy with Mr Boole's function, to represent the result of the elimination of the variables from $d_{x_1} U = 0$, $d_{y_1} U = 0$, $\dots d_{x_n} U = 0$, &c.): this I afterwards found was not the case; and thus I was led to a class of functions, including as a particular case the function θU , all of them possessed of the same characteristic property. The system of partial differential equations was without difficulty replaced by a more fundamental system of equations, upon which, assumed as definitions, the theory appears to me naturally to depend; and it is this view of it which I intend partially to develop in the present paper.

I have already employed the notation

$$\left| \begin{array}{cccc} \alpha, & \beta, & \gamma, & \delta, \dots \\ \alpha', & \beta', & \gamma', & \delta', \dots \\ \alpha'', & \beta'', & \gamma'', & \delta'', \dots \\ & & & \vdots \end{array} \right| \quad (1)$$

(where the number of horizontal rows is less than that of vertical ones) to denote the series of determinants

$$\left| \begin{array}{ccc} \alpha, & \beta, & \gamma, \dots \\ \alpha', & \beta', & \gamma', \dots \\ \alpha'', & \beta'', & \gamma'', \dots \\ & & \vdots \end{array} \right|, \quad (2)$$

which can be formed out of the above quantities by selecting any system of vertical rows; these different determinants not being connected together by the sign $+$, or in any other manner, but being looked upon as perfectly separate.

The fundamental theorem for the multiplication of determinants gives, applied to these, the formula

$$\left| \begin{array}{cccc} A, & B, & C, & D, \dots \\ A', & B', & C', & D', \dots \\ A'', & B'', & C'', & D'', \dots \\ & & & \vdots \end{array} \right| = E \left| \begin{array}{cccc} \alpha, & \beta, & \gamma, & \delta, \dots \\ \alpha', & \beta', & \gamma', & \delta', \dots \\ \alpha'', & \beta'', & \gamma'', & \delta'', \dots \\ & & & \vdots \end{array} \right|, \quad (3)$$

where

$$\left. \begin{array}{l} A = \lambda\alpha + \lambda'\alpha' + \lambda''\alpha'' + \dots \\ B = \lambda\beta + \lambda'\beta' + \lambda''\beta'' + \dots \\ A' = \mu\alpha + \mu'\alpha' + \mu''\alpha'' + \dots \\ B' = \mu\beta + \mu'\beta' + \mu''\beta'' + \dots \\ \vdots \end{array} \right\}, \quad (4)$$

$$E = \left| \begin{array}{cc} \lambda, & \mu, \dots \\ \lambda', & \mu', \dots \\ & \vdots \end{array} \right|, \quad (5)$$

¹The value of a was left undetermined, but Mr Boole has since informed me, he was acquainted with it at the time his paper was written; and has given it in a subsequent paper.

and the meaning of the equation is, that the terms on the first side are equal, each to each, to the terms on the second side.

This preliminary theorem being explained, consider a set of arbitrary coefficients, represented by the general formula

$$\alpha_{rs\dots}, \quad (6)$$

in which the number of symbolical letters $r, s, \dots$ is n , and where each of these is supposed to assume all integer values, from 1 to m inclusively.

Let

$$\left| \begin{array}{ccc} 1 s'_1 t'_1 \dots, & 1 s''_1 t''_1 \dots, & \dots \\ 2 s'_2 t'_2 \dots, & 2 s''_2 t''_2 \dots, & \dots \\ & \vdots & \end{array} \right| \quad (7)$$

represent the whole series, taken in any order, in which the first symbolical letter is α . Similarly,

$$\left| \begin{array}{ccc} r_1 1 t'_1 \dots, & r_1 1 t''_1 \dots, & \dots \\ r_2 2 t'_2 \dots, & r_2 2 t''_2 \dots, & \dots \\ & \vdots & \end{array} \right|, \quad (8)$$

the whole series of those in which the second symbolical letter is α , and so on.

Imagine a function u of the coefficients, which is simultaneously of the forms

$$\left. \begin{array}{l} u = H_p \left| \begin{array}{ccc} 1 s'_1 t'_1 \dots, & 1 s''_1 t''_1 \dots, & \dots \\ 2 s'_2 t'_2 \dots, & 2 s''_2 t''_2 \dots, & \dots \\ & \vdots & \end{array} \right| \\ u = H_p \left| \begin{array}{ccc} r_1 1 t'_1 \dots, & r_1 1 t''_1 \dots, & \dots \\ r_2 2 t'_2 \dots, & r_2 2 t''_2 \dots, & \dots \\ & \vdots & \end{array} \right| \end{array} \right\}; \quad (A)$$

&c.; in which H_p denotes a rational homogeneous function of the order p . The function H is not necessarily supposed to be the same in the above equations, and in point of fact it will not in general be so. The number of equations is of course $= n$.

The function u , whose properties we proceed to investigate, may conveniently be named a “Hyperdeterminant.” Any function satisfying any of the equations (A), without satisfying all of them, will be an “Incomplete Hyperdeterminant.” But, considering in the first place such as are complete—

Let $\overline{rst} \dots$ be a new set of coefficients connected with the former ones by a system of equations of the form

$$\overline{rst} \dots = \lambda_r^1 1st \dots + \lambda_r^2 2st \dots + \dots + \lambda_r^m mst \dots, \quad (9)$$

(where the r in $\lambda_r^1 \dots$ is not an exponent, but an affix).

Suppose $\bar{u}$ is the same function of these new coefficients that u was of the former ones. Then consider the first of the equations (A) and the equation (3), and writing

$$L = \left| \begin{array}{ccc} \lambda_1^1, & \lambda_2^1, & \dots \\ \lambda_1^2, & \lambda_2^2, & \dots \\ & \vdots & \end{array} \right|, \quad (10)$$

we have immediately the equation

$$\bar{u} = L^p \cdot u. \quad (11)$$

Consider the new set of coefficients

$$\overline{rst} \dots = \mu_s^1 r 1 t \dots + \mu_s^2 r 2 t \dots + \dots + \mu_s^m r m t \dots, \quad (12)$$

and $\bar{u}$ the analogous function of these; then, from the second of the equations (A) and the equation (3), and writing

$$M = \left| \begin{array}{ccc} \mu_1^1, & \mu_2^1, & \dots \\ \mu_1^2, & \mu_2^2, & \dots \\ & \vdots & \end{array} \right|, \quad (13)$$

$$\bar{u} = M^p \cdot u = L^p M^p \cdot u. \quad (14)$$

In like manner, considering the new coefficients $\overline{rst} \dots$, where

$$\overline{rst} \dots = \nu_t^1 r s 1 \dots + \nu_t^2 r s 2 \dots + \dots + \nu_t^m r s m \dots, \quad (15)$$

the new function $\bar{u}$, and the quantity N given by

$$N = \begin{vmatrix} \nu_1^1, & \nu_2^1, & \dots \\ \nu_1^2, & \nu_2^2, & \dots \\ \vdots & & \end{vmatrix}, \quad (16)$$

we have, as before,

$$\bar{u} = N^p \cdot u = L^p M^p N^p \cdot u, \quad (17)$$

or

$$\bar{u} = L^p M^p N^p \cdot u; \quad (18)$$

whence, generally, denoting the last result by u' ,

$$u' = L^p M^p N^p \dots u. \quad (\text{B, } 1)$$

Consider now the function

$$\Sigma \Sigma \Sigma \dots (rst \dots x_r y_s z_t \dots), \quad (19)$$

where the Σ 's refer successively to $r, s, t, \dots$, and denote summations from 1 to m inclusively. If u be looked upon as a derivative from the above function, we may write

$$u = \Theta \cdot \Sigma \Sigma \Sigma \dots (rst \dots x_r y_s z_t \dots). \quad (20)$$

Assume

$$\left. \begin{aligned} \bar{x}_r &= \lambda_r^1 x_1 + \lambda_r^2 x_2 + \dots + \lambda_r^m x_m, \\ \bar{y}_s &= \mu_s^1 y_1 + \mu_s^2 y_2 + \dots + \mu_s^m y_m, \\ &\vdots \end{aligned} \right\}. \quad (21)$$

It is easy to obtain

$$\Sigma \Sigma \Sigma \dots (\overline{rst} \dots x_r y_s z_t \dots) = \Sigma \Sigma \Sigma \dots (rst \dots \bar{x}_r \bar{y}_s \bar{z}_t \dots), \dots = \Sigma \Sigma \Sigma \dots (rst \dots \bar{x}_r \bar{y}_s \bar{z}_t \dots), \quad (22)$$

and the formula for (u) becomes

$$\Theta \cdot \Sigma \Sigma \Sigma \dots (\overline{rst} \dots x_r y_s z_t \dots) = L^p M^p N^p \dots \Theta \cdot \Sigma \Sigma \Sigma \dots (rst \dots \bar{x}_r \bar{y}_s \bar{z}_t \dots). \quad (\text{B, } 2)$$

Proceeding to obtain the expression of the coefficients $\overline{rst} \dots$ in terms of the coefficients $rst \dots$, we have

$$\overline{rst} \dots = \Sigma \Sigma \Sigma \dots (\lambda_r^f \mu_s^g \nu_t^h \dots fgh \dots), \quad (\text{C})$$

where the Σ 's refer successively to $f, g, h, \dots$, denoting summations from 1 to m inclusively. Having this equation, it is perhaps as well to retain

$$u' = L^p M^p N^p \dots u, \quad (\text{B, } 1)$$

instead of (B, 2), that form being principally useful in showing the relation of the function u to the theory of the transformation of functions.

It may immediately be seen, that in the equations (B), (C) we may, if we please, omit any number of the marks of variation $()$, omitting at the same time the corresponding signs Σ , and the corresponding factors of the series $L, M, N \dots$.

Also, if u be such as only to satisfy some of the equations (A); then, if in the same formulae we omit the corresponding marks $()$, summatory signs, and terms of the series $L, M, N \dots$, the resulting equations are still true.

From the formulae (A) we may obtain the partial differential equations

$$\Sigma\Sigma\dots\left(\alpha st\dots\frac{d}{d\beta st\dots}\right)u=0, \text{ or } pu, \quad (D)$$

$$\Sigma\Sigma\dots\left(r\alpha t\dots\frac{d}{dr\beta t\dots}\right)u=0, \text{ or } pu,$$

according as α is not equal, or is equal, to β ; and so on: the summatory signs referring in every case to those of the series $r, s, t, \dots$, which are left variable, and extending from 1 to m inclusively.

To demonstrate this, consider the general form of u , as given by the first of the equations (A). This is evidently composed of a series of terms, each of the form

$$cPQR\dots(p \text{ factors}),$$

in which

$$P = \begin{vmatrix} 1 s'_1 t'_1 \dots, & 1 s''_1 t''_1 \dots, & \dots (m \text{ terms}) \\ \alpha s'_\alpha t'_\alpha \dots, & \alpha s''_\alpha t''_\alpha \dots, & \dots \\ \beta s'_\beta t'_\beta \dots, & \beta s''_\beta t''_\beta \dots, & \dots \\ & \vdots & \end{vmatrix}$$

$Q, R, \&c.$ being of the same form; and we have

$$\Sigma\Sigma\dots\left(\alpha st\dots\frac{d}{d\beta st\dots}\right)cPQR\dots = c(QR\dots F + PR\dots Q' + \&c.) + \&c. + \&c.,$$

and

$$\Sigma\Sigma\dots\left(\alpha st\dots\frac{d}{d\beta st\dots}\right)F = \begin{vmatrix} 1 s'_1 t'_1 \dots, & 1 s''_1 t''_1 \dots, & \dots (m \text{ terms}) \\ \alpha s'_\alpha t'_\alpha \dots, & \alpha s''_\alpha t''_\alpha \dots, & \dots \\ \alpha s'_\beta t'_\beta \dots, & \alpha s''_\beta t''_\beta \dots, & \dots \\ & \vdots & \end{vmatrix} = 0;$$

so that all the terms on the second side of the equation vanish. If, however, $\beta = \alpha$, whence, on the second side, we have

$$cQR\dots P + cPR\dots Q + \&c. = p \cdot cPQR\dots + \&c. + \&c. = pU,$$

or the theorem in question is proved.

In the case of an incomplete hyperdeterminant, the corresponding systems of equations are of course to be omitted. In every case it is from these equations that the form of the function u is to be investigated; they entirely replace the system (A).

A very important case of the general theory is, when we suppose the coefficients $rst\dots$ to have the property $r's't' \dots = r''s''t'' \dots$, whenever $r's't' \dots$ and $r''s''t'' \dots$ denote the same combination of letters; and also that the coefficients λ are equal to the coefficients $\mu, \nu \dots$, each to each. In this case the coefficients $\overline{rst} \dots$ have likewise the same property, viz. that $\overline{r''s''t''} \dots = \overline{r's't'} \dots$, whenever $r's't' \dots$ and $r''s''t'' \dots$ denote the same combination of letters.

The equations (B, 1), (B, 2) become in this case

$$u' = L^{\rho p} \cdot u, \quad (B, 3)$$

$$\Theta \cdot \Sigma\Sigma\Sigma\dots \left[\frac{[\alpha]^\rho}{[\alpha']^\rho [\beta']^\rho \dots} \overline{rst} \dots \bar{x}_r \bar{y}_s \bar{z}_t \dots \right] = L^{\rho p} \Theta \cdot \Sigma\Sigma\Sigma\dots \left[\frac{[\alpha]^\rho}{[\alpha']^\rho [\beta']^\rho \dots} rst \dots x_r y_s z_t \dots \right], \quad (B, 4)$$

where only different combinations of values are to be taken for $r, s, t, \dots$ and $\alpha, \beta, \dots$ express how often the same number occurs in the series. In the equation (C), μ, ν must be replaced by λ , the equations (D) are no longer satisfied; the equations (A) reduce themselves to a single one, (so that there can be no question here of incomplete hyperdeterminants): but this is no longer sufficient to determine the function sought after. For this reason, the particular case, treated separately, would be far more difficult than the general one; but the formulae of the general case being first established, these apply immediately to the particular one². The case in question may be defined as that of symmetrical hyperdeterminants, (a denomination

²See concluding paragraph of this paper.

already adopted for ordinary determinants). It would be easily seen what on the same principle would be meant by partially symmetrical hyperdeterminants.

I have not yet succeeded in obtaining the general expression of a hyperdeterminant; the only cases in which I can do so are the following: I. $p = 1$, n even, (if n be odd, there only exist incomplete hyperdeterminants). II. $p = 2$, $m = 2$, n even. III. $p = 3$, $m = 2$, $n = 4$.

I. The first case is, in fact, that of the functions considered at the termination of a paper in the Cambridge Philosophical Transactions, vol. viii. part i. [12]; though at that time I was quite unacquainted with the general theory.

Using the notation there employed, we have

$$u = \left\{ \begin{array}{c} 11 \dots (n) \\ 22 \\ \vdots \\ mm \end{array} \right\}^+,$$

a complete hyperdeterminant when n is even; and when n is odd the functions

$$\left\{ \begin{array}{c} 11 \dots (n) \\ 22 \\ \vdots \\ mm \end{array} \right\}^+, \quad \left\{ \begin{array}{c} 11 \dots (n) \\ 22 \\ \vdots \\ mm \end{array} \right\}^-$$

are each of them incomplete hyperdeterminants.

(A) In the case of $n = 2$, the complete hyperdeterminant is simply the ordinary determinant

$$\begin{vmatrix} 11, & 12, & \dots & 1m \\ 21, & 22, & \dots & 2m \\ & & \vdots & \\ m1, & m2, & \dots & mm \end{vmatrix}.$$

Stating the general conclusion as applied to this case, which is a very well known one,

“If the function

$$U = \Sigma \Sigma (rs \cdot x_r y_s)$$

be transformed into a similar function

$$\Sigma \Sigma (\bar{r}\bar{s} \cdot \bar{x}_r \bar{y}_s).$$

by means of the substitutions

$$\bar{x}_r = \lambda_r^1 x_1 + \lambda_r^2 x_2 + \dots + \lambda_r^m x_m,$$

$$\bar{y}_s = \mu_s^1 y_1 + \mu_s^2 y_2 + \dots + \mu_s^m y_m;$$

then

$$\left| \begin{array}{ccc} \overline{11}, & \overline{12}, & \dots \\ \overline{21}, & \overline{22}, & \\ & & \vdots \end{array} \right| = \left| \begin{array}{ccc} \lambda_1', & \lambda_2', & \dots \\ \lambda_1'', & \lambda_2'', & \dots \\ & & \vdots \end{array} \right| \left| \begin{array}{ccc} \mu_1', & \mu_2', & \dots \\ \mu_1'', & \mu_2'', & \dots \\ & & \vdots \end{array} \right| \left| \begin{array}{ccc} 11, & 12, & \dots \\ 21, & 22, & \\ & & \vdots \end{array} \right|.$$

”

Also, by what has preceded,

$$H = \Sigma \Sigma (\Theta \cdot fg \cdot \lambda_r^f \mu_s^g);$$

so that the theorem is easily seen to amount to the following one—“If the terms of a determinant of the m^{th} order be of the form $\Sigma_r \Sigma_s (rs \cdot x_r \mu_s)$, r, s extending as before, from 1 to m inclusively, the determinant itself is the product of three determinants; the first formed with the coefficients rs , the second with the quantities x , and the third with the quantities y .”

In a following number of the Journal I shall prove, and apply to the theories of Maxima and Minima and of Spherical Coordinates, (I may just mention having obtained, in an elegant form, the formulae for transforming from one oblique set of coordinates to another oblique one) the more general theorem,

“If k be the order of the determinant formed as above, the determinant itself is a quadratic function, its coefficients being determinants formed with the coefficients rs , its variables being determinants formed respectively with the variables x and the variables y ; and the number of variables in each set being the number of combinations of k things out of m , ($= 1$ if $k = m$; if $k > m$ the determinant vanishes).”

I shall give in the same paper the demonstration of a very beautiful theorem, rather relating, however, to determinants than to quadratic functions, proved by Hesse in a Memoir in Crelle’s Journal, vol. xx., “De curvis et superficiebus secundi ordinis;” and from which he has deduced the most interesting geometrical results. Another Memoir, by the same author, Crelle, vol. XXVIII., “Ueber die Elimination der Variabeln aus drei algebraischen Gleichungen vom zweiter Grade mit zwei Variabeln,” though relating in point of fact rather to functions of the third order, contains some most important results. A few theorems on quadratic functions, belonging, however, to a different part of the subject, will be found in my paper already quoted in the Cambridge Philosophical Transactions [12]; and likewise in a paper in the Journal, Chapters in the Algebraical Geometry of n dimensions [11].

I shall, just before concluding this case, write down the particular formula corresponding to three variables, and for the symmetrical case. It is, as is well known, the theorem,

“If

$$U = Ax^2 + By^2 + Cz^2 + 2Fyz + 2Gxz + 2Hxy$$

be transformed into

$$\mathfrak{A}\xi^2 + \mathfrak{B}\eta^2 + \mathfrak{C}\zeta^2 + 2\mathfrak{F}\eta\zeta + 2\mathfrak{G}\xi\zeta + 2\mathfrak{H}\xi\eta$$

by means of

$$\begin{aligned} x &= \alpha\xi + \beta\eta + \gamma\zeta, \\ y &= \alpha'\xi + \beta'\eta + \gamma'\zeta, \\ z &= \alpha''\xi + \beta''\eta + \gamma''\zeta, \end{aligned}$$

then

$$(\mathfrak{A}\mathfrak{B}\mathfrak{C} - \mathfrak{A}\mathfrak{F}^2 - \mathfrak{B}\mathfrak{G}^2 - \mathfrak{C}\mathfrak{H}^2 + 2\mathfrak{F}\mathfrak{G}\mathfrak{H}) =$$

$$(\alpha\beta'\gamma'' - \alpha\beta''\gamma' + \alpha'\beta''\gamma - \alpha'\beta\gamma'' + \alpha''\beta\gamma' - \alpha''\beta'\gamma)^2 (ABC - AF^2 - BG^2 - CH^2 + 2FGH).''$$

(B) Let $n = 3$, and for greater simplicity $m = 2$; write

$$\begin{aligned} a &= 111, & e &= 112, \\ b &= 211, & f &= 212, \\ c &= 121, & g &= 122, \\ d &= 221, & h &= 222, \end{aligned}$$

so that

$$U = a x_1 y_1 z_1 + b x_2 y_1 z_1 + c x_1 y_2 z_1 + d x_2 y_2 z_1 + e x_1 y_1 z_2 + f x_2 y_1 z_2 + g x_1 y_2 z_2 + h x_2 y_2 z_2.$$

There is no complete hyperdeterminant (i.e. for $p = 1$), and the incomplete ones are

$$\begin{aligned} ah - bg - cf + de &= u_1, \text{ suppose,} \\ ah - de - bg + cf &= u_2, \\ ah - cf - de + bg &= u_3. \end{aligned}$$

Thus, suppose the transforming equations are

$$\begin{aligned} \bar{x}_1 &= \lambda_1' x_1 + \lambda_1'' x_2, \\ \bar{x}_2 &= \lambda_2' x_1 + \lambda_2'' x_2; \\ \bar{y}_1 &= \mu_1' y_1 + \mu_1'' y_2, \\ \bar{y}_2 &= \mu_2' y_1 + \mu_2'' y_2; \\ \bar{z}_1 &= \nu_1' z_1 + \nu_1'' z_2, \\ \bar{z}_2 &= \nu_2' z_1 + \nu_2'' z_2; \end{aligned}$$

then

$$\begin{aligned}\bar{u}_1 &= (\mu'_1\mu''_2 - \mu'_2\mu''_1)(\nu'_1\nu''_2 - \nu'_2\nu''_1)u_1, \text{ where } y, z \text{ are changed,} \\ \bar{u}_2 &= (\nu'_1\nu''_2 - \nu'_2\nu''_1)(\lambda'_1\lambda''_2 - \lambda'_2\lambda''_1)u_2, \quad " \quad z, x \quad " \\ \bar{u}_3 &= (\lambda'_1\lambda''_2 - \lambda'_2\lambda''_1)(\mu'_1\mu''_2 - \mu'_2\mu''_1)u_3, \quad " \quad x, y \quad " \end{aligned}$$

We might also have assumed

$$\begin{aligned}u_1 &= ad - bc, \text{ or } eh - gf, \\ u_2 &= af - be, \text{ or } ch - dg, \\ u_3 &= ag - ce, \text{ or } bh - df, \end{aligned}$$

but these are ordinary determinants.

(C) $n = 4, m = 2$.

$$\begin{aligned}a &= 1111, & i &= 1112, \\ b &= 2111, & j &= 2112, \\ c &= 1211, & k &= 1212, \\ d &= 2211, & l &= 2212, \\ e &= 1121, & m &= 1122, \\ f &= 2121, & n &= 2122, \\ g &= 1221, & o &= 1222, \\ h &= 2221, & p &= 2222. \end{aligned}$$

$$\begin{aligned}U &= a x_1 y_1 z_1 w_1 + b x_2 y_1 z_1 w_1 + c x_1 y_2 z_1 w_1 + d x_2 y_2 z_1 w_1 \\ &\quad + e x_1 y_1 z_2 w_1 + f x_2 y_1 z_2 w_1 + g x_1 y_2 z_2 w_1 + h x_2 y_2 z_2 w_1 \\ &\quad + i x_1 y_1 z_1 w_2 + j x_2 y_1 z_1 w_2 + k x_1 y_2 z_1 w_2 + l x_2 y_2 z_1 w_2 \\ &\quad + m x_1 y_1 z_2 w_2 + n x_2 y_1 z_2 w_2 + o x_1 y_2 z_2 w_2 + p x_2 y_2 z_2 w_2, \end{aligned}$$

we have

$$u = ap - bo - cn + dm - el + fk + gj - hi;$$

so that, with the same sets of transforming equations as above, and the additional one,

$$\begin{aligned}\bar{w}_1 &= \rho'_1 w_1 + \rho''_1 w_2, \\ \bar{w}_2 &= \rho'_2 w_1 + \rho''_2 w_2, \end{aligned}$$

we have

$$\bar{u} = (\lambda'_1\lambda''_2 - \lambda'_2\lambda''_1)(\mu'_1\mu''_2 - \mu'_2\mu''_1)(\nu'_1\nu''_2 - \nu'_2\nu''_1)(\rho'_1\rho''_2 - \rho'_2\rho''_1) \cdot u;$$

this is important when viewed in reference to a result which will presently be obtained.

If we take the symmetrical case, we have

$$U = \alpha x^4 + 4\beta x^3 y + 6\gamma x^2 y^2 + 4\delta x y^3 + \epsilon y^4,$$

which is transformed into

$$U' = \alpha' x'^4 + 4\beta' x'^3 y' + 6\gamma' x'^2 y'^2 + 4\delta' x' y'^3 + \epsilon' y'^4,$$

by means of

$$\begin{aligned}x &= \lambda x' + \mu y', \\ y &= \lambda' x' + \mu' y'; \end{aligned}$$

if

$$\begin{aligned}u &= \alpha\epsilon - 4\beta\delta + 3\gamma^2, \\ u' &= \alpha'\epsilon' - 4\beta'\delta' + 3\gamma'^2, \end{aligned}$$

we have

$$u' = (\lambda\mu' - \lambda'\mu)^4 \cdot u.$$

II. Where $p = 2, m = 2, n$ is odd.

The expression

$$u = \left| \begin{array}{c} (1111 \dots (n)) \\ 1222 \\ 1111 \dots (n) \\ 2222 \end{array} \right|^2 + \left| \begin{array}{c} (1111 \dots (n)) \\ 2222 \\ 2111 \dots \\ 2222 \end{array} \right|^2$$

is a complete hyperdeterminant; and that over whichever row the mark ($\dagger$) of non-permutation is placed. The different expressions so obtained are not, however, all of them independent functions: thus, in the following example, where $n = 3$, the three functions are absolutely identical.

(A). $n = 3$, notation as in I. (B).

$$u = a^2 h^2 + b^2 g^2 + c^2 f^2 + d^2 e^2 - 2ahbg - 2ahcf - 2ahde - 2bgcf - 2bgde - 2cfde + 4adfg + 4bech,$$

and then

$$\bar{u} = (\lambda'_1 \lambda''_2 - \lambda'_2 \lambda''_1)^2 (\mu'_1 \mu''_2 - \mu'_2 \mu''_1)^2 (\nu'_1 \nu''_2 - \nu'_2 \nu''_1)^2 u.$$

This is in many respects an interesting example. We see that the function u may be expressed in the three following forms:

$$u = (ah - bg - cf + de)^2 + 4(ad - bc)(fg - eh), \quad (1)$$

$$u = (ah - bg - de + cf)^2 + 4(af - be)(dg - ch), \quad (2)$$

$$u = (ah - cf - de + bg)^2 + 4(ag - ce)(df - bh), \quad (3)$$

which are indeed the direct results of the general form above given, the sign ($\dagger$) being placed in succession over the different columns: and the three forms, as just remarked, are in this case identical.

We see from the first of these that u is of the second or third, from the second that u is of the first or third, from the third that u is of the first or second of the three following forms:

$$u = H_2 \left| \begin{array}{c} a, b, c, d \\ e, f, g, h \end{array} \right|, \quad u = H_2 \left| \begin{array}{c} a, b, e, f \\ c, d, g, h \end{array} \right|, \quad u = H_2 \left| \begin{array}{c} a, c, e, g \\ b, d, f, h \end{array} \right|,$$

which is as it should be.

The following is a singular property of u :

Let

$$a' = \frac{1}{2} \frac{du}{da}, \quad b' = \frac{1}{2} \frac{du}{db}, \quad \dots, \quad h' = \frac{1}{2} \frac{du}{dh};$$

then, u' being the same function of these new coefficients that u is of the former ones,

$$u' = u^3.$$

To prove this, write

$$p = ah - bg - cf + de, \quad q = (ad - bc), \quad r = eh - fg;$$

$$a_1 = ap - 2qe, \quad e_1 = -2ra + pe,$$

$$b_1 = bp - 2qf, \quad f_1 = -2rb + pf,$$

$$c_1 = cp - 2qg, \quad g_1 = -2rc + pg,$$

$$d_1 = dp - 2qh, \quad h_1 = -2rd + ph;$$

we have, as a particular case of the general formula just obtained,

$$u_1 = (p^2 - 4qr)^2 u = u_1^2 \cdot u, = u^3.$$

Also

$$\begin{aligned} a_1 &= h', & e_1 &= d', \\ b_1 &= -g', & f_1 &= -c', \\ c_1 &= -f', & g_1 &= -b', \\ d_1 &= e', & h_1 &= a'; \end{aligned}$$

whence $u_1 = u'$, that is $u' = u^3$.

There is no difficulty in showing also, that if $a'', b'', \dots h''$ are derived from $a', b', \dots h'$, as these are from $a, b, \dots h$, then

$$a'' = u^2 a, \quad b'' = u^2 b, \quad \dots \quad h'' = u^2 h.$$

The particular case of this theorem, which corresponds to symmetrical values of the coefficients, is given by M. Eisenstein, Crelle, vol. xxvii. [1844], as a corollary to his researches on the cubic forms of numbers.

Considering this symmetrical case

$$\begin{aligned} U &= \alpha x^3 + 3\beta x^2 y + 3\gamma x y^2 + \delta y^3, \\ u &= \alpha^2 \delta^2 - 6\alpha \delta \beta \gamma - 3\beta^2 \gamma^2 + 4\beta^3 \delta + 4\alpha \gamma^3, \end{aligned}$$

so that if U be transformed into

$$U' = \alpha' x'^3 + 3\beta' x'^2 y' + 3\gamma' x' y'^2 + \delta' y'^3,$$

by means of

$$\begin{aligned} x &= \lambda x' + \mu y', \\ y &= \lambda' x' + \mu' y', \end{aligned}$$

then if

$$u' = \alpha'^2 \delta'^2 - 6\alpha' \delta' \beta' \gamma' - 3\beta'^2 \gamma'^2 + 4\beta'^3 \delta' + 4\alpha' \gamma'^3,$$

we have

$$u' = (\lambda \mu' - \lambda' \mu)^6 \cdot u.$$

III. $p = 3, m = 2, n = 4$.

Notation as in I. (C),

$$u = A(\mathfrak{A} + 3\mathfrak{B} + 3\mathfrak{C} + 6\mathfrak{D} + 6\mathfrak{E}) - B(\mathfrak{F} + \mathfrak{G} - 3\mathfrak{H} - 3\mathfrak{I} + 2\mathfrak{J} + 3\mathfrak{K}),$$

where A, B are arbitrary constants, and $\mathfrak{A}, \mathfrak{B}, \&c. \dots \mathfrak{K}$ are functions of the coefficients, given as follows:—

$$\begin{aligned} \mathfrak{A} &= -a^2 p^2 i o + b^2 o^2 a p + c^2 n^2 d m - d^2 m^2 h i + e^2 l^2 f k - f^2 k^2 e l - g^2 j^2 h i + h^2 i^2 g j \\ &\quad - a^2 p^2 c n + b^2 o^2 d m + c^2 n^2 a p - d^2 m^2 b o + e^2 l^2 g j - f^2 k^2 h i - g^2 j^2 e l + h^2 i^2 f k \\ &\quad - a^2 p^2 e l + b^2 o^2 f k + c^2 n^2 g j - d^2 m^2 h i + e^2 l^2 a p - f^2 k^2 b o - g^2 j^2 c n + h^2 i^2 d m \\ &\quad - a^2 p^2 h i + b^2 o^2 g j + c^2 n^2 f k - d^2 m^2 e l + e^2 l^2 d m - f^2 k^2 c n - g^2 j^2 b o + h^2 i^2 a p. \end{aligned}$$

$$\begin{aligned} \mathfrak{B} &= +a^2 p^2 d m - b^2 o^2 c n - c^2 n^2 b o + d^2 m^2 a p - e^2 l^2 h i + f^2 k^2 g j + g^2 j^2 f k - h^2 i^2 e l \\ &\quad + a^2 p^2 f k - b^2 o^2 e l - c^2 n^2 h i + d^2 m^2 g j - e^2 l^2 b o + f^2 k^2 a p + g^2 j^2 d m - h^2 i^2 c n \\ &\quad + a^2 p^2 g j - b^2 o^2 h i - c^2 n^2 e l + d^2 m^2 f k - e^2 l^2 c n + f^2 k^2 d m + g^2 j^2 a p - h^2 i^2 b o. \end{aligned}$$

$$\begin{aligned} \mathfrak{C} &= a p b o c n - a p b o d m - a p c n d m + b o c n d m - e l f k g j + e l f k h i + e l g j h i - h i g j f k \\ &\quad + a p b o e l - a p b o j k - c n d m g j + d m c n h i - e l f k a p + e l f k b o + g j h i c n - h i g j d m \\ &\quad - a p b o g j + a p b o h i + c n d m e l - d m c n k f + e l f k c n - e l f k d m - g j h i a p + h i g j b o \\ &\quad + a p c n e l - b o d m f k - c n a p g j + d m b o k i - e l g j a p + f k h i b o + g j e l c n - h i f k d m \\ &\quad - a p c n f k + b o d m e l + c n a p h i - d m b o e l + e l g j b o - f k h i a p - g j e l d m + h i f k c n \\ &\quad - a p d m e l + b o c n k f + c m b o g j - d m a p h i + e l h i a p - f k g j b o - g j f k c n + h i d m e l. \end{aligned}$$

$$\mathfrak{D} = a p d m j k - b o c n e l - b o c n i k + d m a p g j - e l h i b o + f k a p g i + g j f k d m - h i e l c n.$$

$$\begin{aligned}\mathfrak{E} = & a^2phjo - b^2goip - c^2hifn + d^2emkn - e^2dlkn + f^2ckml + g^2bjpi - h^2aijo \\ & - i^2phbg + j^2ogah + k^2nfde - l^2mecf + m^2ldcf - n^2ckde - o^2bjah + p^2aibg \\ & + a^2phkn - b^2golm - c^2nfip + d^2emjo - e^2dljo + f^2ckip + g^2bjpl - h^2imk \\ & - i^2phcf + j^2ogde + k^2nfah - l^2mebg + m^2ldbg - n^2ckah - o^2bjde + p^2aicf \\ & + a^2phlm - b^2gokn - c^2hifjo + d^2emip - e^2dlpi + f^2ckjo + g^2bjnk - h^2aiml \\ & - i^2phde + j^2ogcf + k^2nfbg - l^2meah + m^2ldah - n^2ckbg - o^2bjcf + p^2aide \\ & + a^2pdno - b^2cmpp - c^2bpmn + d^2aomn - e^2hijkl + f^2gilk + g^2flij - h^2ekji \\ & - i^2lfgh + j^2kehg + k^2jhef - l^2igfe + m^2pbcd - n^2aodc - o^2ndab + p^2mbca \\ & + a^2plng - b^2okmo - c^2ejpn + d^2fiom - e^2cpjl + f^2doik + g^2anlj - h^2bmki \\ & - i^2odfh + j^2pceg + k^2mhbf - l^2nage + m^2khbd - n^2lgac - o^2ifbd + p^2heca \\ & + a^2plfo - b^2ekpo - c^2hymn + d^2gimn - e^2bpkl + f^2aolk + g^2dnij - h^2cmji \\ & - i^2hdgh + j^2nchg + k^2pbe f - l^2oafe + m^2jhcd - n^2lgcd - o^2ifab + p^2keba.\end{aligned}$$

$$\begin{aligned}\mathfrak{F} = & apbgkn - boahm - cndiep + dmcjfo \\ & - elfocj + fkpied + gjhinla - higjnhk \\ & - ihjocf + gjidep + kflamh - lekbng \\ & + mdnhkg - ncmhal - obpied + paofjc \\ & + apbglm - boahkn - cndejo + dmcjip \\ & - elcfip + fkedoj + gjhakn - hibgml \\ & - ihjode + gijpcf + kflmbg - lekna h \\ & + mdknah - ncmlbg - obpicf + paojde \\ & + apcflg - bojehk - onjehk + dmilgf \\ & - elmpbc + fkadno + gjadno - hipmcb \\ & - ihadno + gjbmcp + kfc bpm - eladno \\ & + mdjehk - ncilfg - obilfg + pahejk \\ & + apcfhn - bodhie - cnahjo + dmbgip \\ & - elbgip + fkahjo + gjednk - hicfml \\ & - ihknde + mdahjo + kfipbg - bocfml \\ & + gjcfml - ncpigb - Leahjo + paknde \\ & + apidng - bojcmh - cnbkpe + dmaflo \\ & - elmhjc - fkidng + gjaflo - ihbkpe \\ & - iha flo + gjbkpe + fkcjhm - elidng \\ & + mdbkpe - cna flo - boidng + pahmcj \\ & + apidfo - bojcep - cnbkhm + dmialng \\ & - elmnbk - fkalng + gjidfo - ihpecj \\ & - ihalng + gjkbmh + fkcjpe - elidfo \\ & + mdjcep - ncidfo - obalvg + pahmbk.\end{aligned}$$

$$\begin{aligned}\mathfrak{G} = & a^2hord - b^2pgmk - c^2pf nj + d^2onie - e^2dpkj + f^2lkco + g^2lni - h^2amkj \\ & - i^2pdfg + j^2oech + k^2nbeh - l^2mafg + m^2lch - n^2kadg - o^2jadf + p^2ibec : \end{aligned}$$

we have, as usual,

$$\bar{u} = (\lambda'_1\lambda''_2 - \lambda'_2\lambda''_1)^3(\mu'_1\mu''_2 - \mu'_2\mu''_1)^3(\nu'_1\nu''_2 - \nu'_2\nu''_1)^3(\rho'_1\rho''_2 - \rho'_2\rho''_1)^3 \cdot u.$$

Particular forms of U are

$A = 1, B = 0$:

$$u = \mathfrak{A} + 3\mathfrak{B} + 3\mathfrak{C} + 6\mathfrak{D} + 6\mathfrak{E} = (ap - bo - cn + dm - el + fk + gj - hi)^2, = v \text{ suppose.}$$

$A = 1, B = 9$:

$$u = \mathfrak{A} + 3\mathfrak{B} - 6\mathfrak{C} - 3\mathfrak{D} + 3\mathfrak{E}\mathfrak{F} + 9\mathfrak{G} - 18\mathfrak{H} - 27\mathfrak{K}, = \theta U \text{ suppose,}$$

where $\theta U = 0$ is the result of the elimination of the variables from the equations $d_{x_1}U = 0$, $d_{y_1}U = 0$, $d_{z_1}U = 0$, $d_{w_1}U = 0$, $d_{x_2}U = 0$, $d_{y_2}U = 0$, $d_{z_2}U = 0$, $d_{w_2}U = 0$. In fact, by an investigation similar to Mr Boole's, applied to a function such as U , it is shown that θU has the characteristic property of the function u : also in the present case u is the most general function of its kind, so that θU is obtained from U by properly determining the constants. This has been effected by comparing the value of u , in the symmetrical case, with the value of θU , in the same case, the expanded expression of which is given by Mr Boole in the Journal, vol. iv. p. 169. Assuming $A = 1$, the result was $B = 9$. [Incorrect; the result of the elimination is not $\theta U = 0$, but an equation of a higher degree.]

The general form of u now becomes

$$u = \alpha v^2 + \beta \theta U$$

in which α, β , are indeterminate.

We have

$$\bar{u} = \alpha \bar{v}^2 + \beta \theta \bar{U} = M (\alpha v^2 + \beta \theta U),$$

where

$$M = (\lambda'_1 \lambda''_2 - \lambda'_2 \lambda''_1)^4 (\mu'_1 \mu''_2 - \mu'_2 \mu''_1)^4 (\nu'_1 \nu''_2 - \nu'_2 \nu''_1)^4 (\rho'_1 \rho''_2 - \rho'_2 \rho''_1)^4,$$

and thence

$$\bar{v}^2 = M v^2,$$

which coincides with a previous formula, and

$$\theta \bar{U} = M \theta U :$$

whence, eliminating M ,

$$\frac{\theta \bar{U}}{\bar{v}^2} = \frac{\theta U}{v^2}$$

an equation which is remarkable as containing only the constants of U and $\bar{U}$: it is an equation of condition which must exist among the constants of $\bar{U}$ in order that this function may be derivable by linear substitutions from U .

In the symmetrical case, or where

$$U = \alpha x^4 + 4\beta x^3 y + 6\gamma x^2 y^2 + 4\delta x y^3 + \epsilon y^4,$$

it has been already seen that v is given by

$$v = \alpha \epsilon - 4\beta \delta + 3\gamma^2.$$

Proceeding to form θU , we have

$$\begin{aligned} \mathfrak{A} &= \alpha^2 \epsilon^2 - 4\beta^3 \delta^3 + 3\gamma^4, \\ \mathfrak{B} &= 4(\alpha \epsilon^2 \delta^2 - \alpha^2 \epsilon^2 \delta + 3\gamma^2 \beta^2 \delta^2 - 4\beta^3 \delta^3), \\ \mathfrak{C} &= 3(\alpha^2 \epsilon^2 \gamma^2 + 2\gamma^4 + \alpha \epsilon \gamma^4 - 4\beta^2 \delta^2), \\ \mathfrak{D} &= 6(\alpha \epsilon^2 \delta^2 - 2\alpha \epsilon \beta \delta \gamma^2 + 3\beta^2 \gamma^2 \delta^2 - 2\beta \delta \gamma^4), \\ \mathfrak{E} &= 3\alpha \epsilon \gamma^4 - 4\beta^3 \delta^3 + \gamma^4, \\ \mathfrak{F} &= 6(\alpha^2 \gamma \epsilon + \epsilon^2 \beta^2 \gamma - 2\beta^3 \epsilon \gamma \delta - 2\delta^3 \alpha \beta \gamma - 4\beta^2 \gamma^2 \delta^2 + 4\beta \delta \gamma^2 + \gamma^3 \beta^2 \epsilon + \gamma \alpha \delta^2), \\ \mathfrak{G} &= 12(\alpha \epsilon \beta \delta \gamma^2 - \alpha \gamma^3 \delta^2 - \epsilon \gamma^3 \beta^2 + \gamma^2 \alpha \delta^2 + \gamma \epsilon \beta^3 + \beta \delta \gamma^4 - 2\beta^2 \gamma^2 \delta^2), \\ \mathfrak{H} &= (\alpha^2 \delta^4 + \epsilon^2 \beta^4 - 4\beta^3 \epsilon \gamma^3 - 4\alpha \gamma \beta^3 + 6\beta^2 \gamma^2 \delta^2), \end{aligned}$$

and these values give

$$\begin{aligned} \theta U &= \alpha^2 \epsilon^4 - 6\alpha \beta^2 \delta^2 \epsilon - 12\alpha^2 \beta \delta \epsilon^2 - 18\alpha^2 \gamma^2 \epsilon^2 - 27\alpha^2 \delta^4 - 27\beta^4 \epsilon^2 + 36\beta^2 \gamma \delta^2 \epsilon \\ &\quad + 54\alpha \beta \gamma \delta^2 \epsilon + 54\alpha \beta^2 \gamma \epsilon^2 - 54\alpha \gamma^3 \delta^2 - 54\beta^3 \gamma^2 \epsilon - 64\beta^3 \delta^3 + 81\alpha^2 \delta \epsilon \\ &\quad + 108\alpha \gamma \beta \delta^3 + 108\beta^2 \gamma \delta \epsilon - 180\alpha \beta \gamma^2 \delta \epsilon, \end{aligned}$$

so that this function, divided by $(\alpha \epsilon - 4\beta \delta + 3\gamma^2)^3$, is invariable for all functions of the fourth order which can be deduced one from the other by linear substitutions. The function $\alpha \epsilon - 4\beta \delta + 3\gamma^2$ occurs in other

investigations: I have met with it in a problem relating to a homogeneous function of two variables, of any order whatever, $\alpha, \beta, \gamma, \delta, \epsilon$ signifying the fourth differential coefficients of the function. But this is only remotely connected with the present subject.

Since writing the above, Mr Boole has pointed out to me that in the transformation of a function of the fourth order of the form $ax^4 + 4bx^3y + 6cx^2y^2 + 4dxy^3 + ey^4$,— besides his function θu , and my quadratic function $ae - 4bd + 3c^2$,—there exists a function of the third order $ace - b^2e - ad^2 - c^3 + 2bdc$, possessing precisely the same characteristic property, and that, moreover, the function θu may be reduced to the form

$$(ae - 4bd + 3c^2)^3 - 27(ace - ad^2 - eb^2 - c^3 + 2bdc)^2;$$

the latter part of which was verified by trial; the former he has demonstrated in a manner which, though very elegant, does not appear to be the most direct which the theorem admits of. In fact, it may be obtained by a method just hinted at by Mr Boole, in his earliest paper on the subject, *Mathematical Journal*, vol. II. p. 70. The equations $d_x^2u = 0$, $d_xd_yu = 0$, $d_y^2u = 0$, imply the corresponding equations for the transformed function: from these equations we might obtain two relations between the coefficients, which, in the case of a function of the fourth order, are of the orders 3 and 4 respectively: these imply the corresponding relations between the coefficients of the transformed function. Let $A = 0$, $B = 0$, $A' = 0$, $B' = 0$, represent these equations; then, since $A = 0$, $B = 0$, imply $A' = 0$, we must have $A' = \Lambda A + MB$, Λ, M , being functions of $\lambda, \lambda', \mu, \mu'$: but B being of the fourth order, while A, A' are only of the third order in the coefficients of u , it is evident that the term MB must disappear, or that the equation is of the form $A' = \Lambda A$. The function A is obviously the function which, equated to zero, would be the result of the elimination of x^2, xy, y^2 , considered as independent quantities from the equations $ax^2 + 2bxy + cy^2 = 0$, $bx^2 + 2cxy + dy^2 = 0$, $cx^2 + 2dxy + ey^2 = 0$, viz. the function given above. Hence the two functions on which the linear transformation of functions of the fourth order ultimately depend are the very simple ones

$$ae - 4bd + 3c^2, \quad ace - ad^2 - eb^2 - c^3 + 2bdc,$$

the function of the sixth order being merely a derivative from these. The above method may easily be extended: thus for instance, in the transformation of functions of any even order, I am in possession of several of the transforming functions; that of the fourth order, for functions of the sixth order, I have actually expanded: but it does not appear to contain the complete theory. Again, in the particular case of homogeneous functions of two variables, the transforming functions may be expressed as symmetrical functions of the roots of the equation $u = 0$, which gives rise to an entirely distinct theory. This, however, I have not as yet developed sufficiently for publication. There does not appear to be anything very directly analogous to the subject of this note, in my general theory: if this be so, it proves the absolute necessity of a distinct investigation for the present case, that which I have denominated the symmetrical one.

14.

ON LINEAR TRANSFORMATIONS.

[From the Cambridge and Dublin Mathematical Journal, vol. i. (1846), pp. 104–122.]

In continuing my researches on the present subject, I have been led to a new manner of considering the question, which, at the same time that it is much more general, has the advantage of applying directly to the only case which one can possibly hope to develop with any degree of completeness, that of functions of two variables. In fact the question may be proposed, “To find all the derivatives of any number of functions, which have the property of preserving their form unaltered after any linear transformations of the variables.” By Derivative I understand a function deduced in any manner whatever from the given functions, and I give the name of Hyperdeterminant Derivative, or simply of Hyperdeterminant, to those derivatives which have the property just enunciated. These derivatives may easily be expressed explicitly, by means of the known method of the separation of symbols. We thus obtain the most general expression of a hyperdeterminant. But there remains a question to be resolved, which appears to present very great difficulties, that of determining the independent derivatives, and the relation between these and the remaining ones. I have only succeeded in treating a very particular case of this question, which shows however in what way the general problem is to be attacked.

Imagine p series each of m variables

$$x_1, y_1, \dots \&c. \quad x_2, y_2, \dots \&c. \quad \dots \quad x_p, y_p, \dots \&c.,$$

where p is at least as great as m .

Similarly p' series each of m' variables

$$x'_1, y'_1, \dots \&c. \quad x'_2, y'_2, \dots \&c. \quad \dots \quad x'_{p'}, y'_{p'}, \dots \&c.,$$

p' at least as great as m' , and so on. Let the analogous variables $\bar{x}, \bar{y}, \dots$ be connected with these by the equations

$$\begin{aligned} \bar{x} &= \lambda \bar{x} + \mu \bar{y} + \dots, \\ \bar{y} &= \lambda' \bar{x} + \mu' \bar{y} + \dots, \\ \bar{x}' &= \lambda \bar{x}' + \mu \bar{y}' + \dots, \\ \bar{y}' &= \lambda'' \bar{x}' + \mu'' \bar{y}' + \dots, \\ &\vdots \end{aligned}$$

where $x, y, \dots$ stand for $x_1, y_1, \dots$ or $x_2, y_2, \dots$ or $x_p, y_p, \dots$; $x', y', \dots$ stand for $x'_1, y'_1, \dots$ or $x'_2, y'_2, \dots$ or $x'_{p'}, y'_{p'}, \dots$ &c. ... The coefficients $\lambda, \mu, \dots, \lambda', \mu', \dots$ &c.; $\lambda', \mu', \dots, \lambda'', \mu'', \dots$ remain the same in all these systems. Suppose next,

$$\xi = \delta_x, \quad \eta = \delta_y, \dots$$

i.e.

$$\xi_1 = \delta_{x_1}, \quad \eta_1 = \delta_{y_1}, \dots \xi_2 = \delta_{x_2}, \dots$$

(where $\delta_x, \delta_y, \dots$ are the symbols of differentiation relative to x, y , &c.). Then evidently

$$\xi = \lambda \xi + \lambda' \eta + \dots,$$

$$\eta = \mu \xi + \mu' \eta + \dots,$$

with similar equations for $\xi', \eta', \dots$ Suppose

$$\|\Omega\| = \begin{vmatrix} \xi_1 & \xi_2 & \dots & \xi_p \\ \eta_1 & \eta_2 & \dots & \eta_p \\ & & & \vdots \end{vmatrix}, \quad \|\Omega'\| = \begin{vmatrix} \xi'_1 & \xi'_2 & \dots & \xi'_{p'} \\ \eta'_1 & \eta'_2 & \dots & \eta'_{p'} \\ & & & \vdots \end{vmatrix},$$

that is to say $\|\Omega\|$ is the series of determinants formed by choosing any m vertical columns to compose a determinant, and similarly $\|\Omega'\|$, &c. Suppose, besides,

$$E = \begin{vmatrix} \lambda & \mu & \dots \\ \lambda' & \mu' & \dots \\ & & \vdots \end{vmatrix}, \quad E' = \begin{vmatrix} \lambda' & \mu' & \dots \\ \lambda'' & \mu'' & \dots \\ & & \vdots \end{vmatrix}.$$

Then, by the known properties of determinants,

$$\|\bar{\Omega}\| = E \|\Omega\|, \quad \|\bar{\Omega}'\| = E' \|\Omega'\| \&c.,$$

i.e. the terms on the one side are respectively equal to the terms on the other. Hence if

$$\square = F(\|\Omega\|^f, \|\Omega'\|^{f'}, \dots),$$

i.e. $\square$ a rational and integral function, homogeneous of the order f in the quantities of the series $\|\Omega\|$, homogeneous of the order f' in the quantities of the series $\|\Omega'\|$, &c., we have immediately

$$\bar{\square} = E^f E'^{f'} \dots \square;$$

or if U be any function whatever of the variables $x, y, \dots$ which is transformed by the linear substitutions above into $\bar{U}$, then

$$\bar{\square} \bar{U} = E^f E'^{f'} \dots \square U;$$

or the function

$$\square U$$

is by the above definition a hyperdeterminant derivative. The symbol $\square$ may be called “symbol of hyperdeterminant derivation,” or simply “hyperdeterminant symbol.”

Let $A, B, \dots$ represent the different quantities of the series $\|\Omega\|$, $A', B', \dots$ those of the series $\|\Omega'\|$, &c.; then $\square$ may be reduced to a single term, and we may write

$$\square = A^a B^b \dots A'^{a'} B'^{b'} \dots$$

Also U may be supposed of the form

$$U = \Phi \Phi' \dots$$

where Φ, Φ' are functions of the variables of one of the sets $x, y, \dots$, of one of the sets $x'', y'', \dots$, &c., thus Φ is of the form

$$F(x_1, y_1, \dots x_1'', y_1'', \dots),$$

and so on. The functions $\Phi, \Phi' \dots$ may be the same or different. It may be supposed after the differentiations that several of the sets $x, y, \dots$ or of the sets $x', y', \dots$ become identical: in such cases it will always be assumed that the functions $\Phi, \dots$ into which these sets of variables enter, are similar; so that they become absolutely identical, when the variables they contain are made so. Thus the general expression of a hyperdeterminant is

$$\square U = A^a B^b \dots A'^{a'} B'^{b'} \dots \Phi^k \dots$$

in which, after the differentiations, any number of the sets of variables are made equal. For instance, if all the sets $x, y, \dots$ and all the sets $x', y', \dots$ are made equal, the hyperdeterminant refers to a single function $F(x, y \dots x', y' \dots)$. In any other case it refers not to a single function but to several.

What precedes, is the general theory: it might perhaps have been made clearer by confining it to a particular case: and by doing this from the beginning it will be seen that it presents no real difficulties. Passing at present to some developments, to do this, I neglect entirely the sets $x', y' \dots$ and I assume that the number m of variables in each of the sets $x, y \dots$ reduces itself to two; so that I consider functions of two variables x, y only. The functions Φ, Φ' , &c. reduce themselves to functions $V_1, V_2 \dots V_p$ of the variables x_1, y_1 , or $x_2, y_2 \dots$ or x_p, y_p . Writing also

$$\square = 12^\alpha 13^\beta 14^\gamma \dots 23^{\alpha'} 24^{\beta'} \dots 34^{\alpha''} \dots$$

the symbols $A, B \dots$ reduce themselves to $12, 13 \dots$. Hence for functions of two variables, there results the following still tolerably general form

$$\square U = 12^\alpha 13^\beta 14^\gamma \dots 23^{\alpha'} 24^{\beta'} \dots 34^{\alpha''} \dots V_1 V_2 V_3 V_4 \dots$$

The functions $V_1, V_2 \dots$ may be the same or different: but they will be supposed the same whenever the corresponding variables are made equal. This equality will be denoted by writing, for instance,

$$\square V V' V V' \dots$$

to represent the value assumed by

$$\square V_1 V_2 V_3 V_4 \dots$$

when after the differentiations

$$x_1 = x_3, y_1 = y_3, x_2 = x_4, y_2 = y_4;$$

&c.

It is easy to determine the general term of $\square U$. To do this, writing for shortness

$$\alpha + \beta + \gamma \dots = f_1,$$

$$\alpha + \beta' + \gamma' \dots = f_2,$$

$$\beta + \beta' + \gamma'' \dots = f_3,$$

&c.

$$\delta_{x_1}^{f_1-r-s-t\dots} \delta_{y_1}^{r+s+t\dots} V_1 = V_1^{r+s+t\dots} \text{ or } \delta_{x_1}^{r+s+t\dots} \delta_{y_1}^{f_1-r-s-t\dots} V_1 = V_1^{-r-s-t\dots} \text{ or } V_1^{r+s+t\dots},$$

the general term is

$$\pm V_1^{r+s+t\dots} V_2^{\alpha-r+s'+t'\dots} V_3^{\beta-s+\beta'-s'+t''\dots} \dots$$

where $r, s, t, \dots s', t', \dots t'', \dots$ extend from 0 to $\alpha, \beta, \gamma, \dots \beta', \gamma', \dots \gamma'', \dots$ respectively. It would be easy to change this general term in a way similar to that which will be employed presently for the particular case of $\square V_1 V_2 V_3 V_4$.

If several of the functions become identical, and for these some of the letters f are equivalent, it is clear that the derivative $\square U$ refers to a certain number of functions $V_1, V_2 \dots$ the same or different, of the variables $x, y; x', y'; \dots$ and besides that this derivative is homogeneous, of the degrees $\theta_1, \theta'_1, \dots$ with respect to the differential coefficients of the orders $f_1, f'_1, \dots$ &c. of V_1 , (consequently homogeneous of the order $\theta_1 + \theta'_1 + \dots$ with respect to these differential coefficients collectively), homogeneous and of the degrees $\theta_2, \theta'_2, \dots$ with respect to the differential coefficients of the orders $f_2, f'_2 \dots$ of V_2 , (consequently of the order $\theta_2 + \theta'_2 \dots$ with respect to these collectively), and so on. The degree with respect to all the functions is of course $\theta_1 + \theta'_1 \dots + \theta_2 + \theta'_2 + \dots = p$ suppose. In general, only a single function will be considered, and it will be assumed that $\square U$ only contains the differential coefficients of the f^{th} order. In this case, the derivative is said to be of the degree p and of the order f . The most convenient classification is by degrees, rather than by orders.

Commencing with the simplest case, that of functions of the second order (and writing V, W instead of V_1, V_2), we have

$$\square VW = 12^\alpha VW,$$

(where ξ_1, η_1 apply to V and ξ_2, η_2 to W). This will be constantly represented in the sequel by the notation

$$12^\alpha VW = B_\alpha(V, W).$$

Hence, writing

$$\delta_x^a V = V^{,a}, \quad \delta_x^{a-1} \delta_y V = V^{,a-1,1}, \dots,$$

we have

$$B_\alpha(V, W) = V^{,a} W^a, - \binom{a}{1} V^{,a-1,1} W^{a-1,1} + \dots;$$

and in particular, according as a is odd or even,

$$B_a(V, V) = 0,$$

$$B_a(V, V) = V^{,a} V^a, - V^{a-1,1} V^{,a-1,1} + \dots,$$

continued to the term which contains $V^{\frac{a}{2}, \frac{a}{2}} V^{, \frac{a}{2}, \frac{a}{2}}$, the coefficient of this last term being divided by two.

Thus, for the functions $\frac{1}{2}(ax^2 + 2bxy + cy^2)$, $\frac{1}{4}(ax^4 + 4bx^3y + 6cx^2y^2 + 4dxy^3 + ey^4)$, &c., if a be made equal to 2, 4, &c. respectively, we have the constant derivatives

$$\begin{aligned} ac - b^2, \\ ae - 4bd + 3c^2, \\ ag - 6bf + 15ce - 10d^2, \\ ai - 8bh + 28cg - 56df + 35e^2, \end{aligned}$$

which have all of them the property of remaining unaltered, *à un facteur près*, when the variables are transformed by means of $x = \lambda x' + \mu y'$, $y = \lambda' x' + \mu' y'$. Thus, for instance, if these equations give

$$ax^2 + 2bxy + cy^2 = \bar{a}\bar{x}^2 + 2\bar{b}\bar{x}\bar{y} + \bar{c}\bar{y}^2,$$

then

$$\bar{a}\bar{c} - \bar{b}^2 = (\lambda\mu' - \lambda'\mu)^2 \cdot (ac - b^2).$$

and so on. This is the general property, which we call to mind for the case of these constant derivatives.

The above functions may be transformed by means of the identical equation

$$B_a(V, W) = 12^{a-k} B_k(V, W),$$

to make use of which, it is only necessary to remark the general formula

$$\delta_{x_1}^{a-k} \delta_{x_2}^{a-k} VW = V^{,a-k} W^{,a-k}.$$

Thus, if $k = 1$, we obtain for the above series, the new forms

$$\begin{aligned} & ac - b^2, \\ & (ae - bd) - 3(bd - c^2), \\ & (ag - bf) - 5(bf - ce) + 10(ce - d^2), \\ & \text{\&c.}, \\ & (ai - bh) - 7(bh - cg) + 21(cg - df) - 35(df - e^2), \end{aligned}$$

the law of which is evident. This shows also that these functions may be linearly expressed by means of the series of determinants

$$\begin{vmatrix} a, & b \\ b, & c \end{vmatrix}, \quad \begin{vmatrix} a, & b, & c \\ b, & c, & d \end{vmatrix}, \quad \text{\&c.}$$

We may also immediately deduce from them the derivatives B which relate to two functions. For example, for functions of the sixth order this is

$$ag' + a'g - 6(bf' + b'f) + 15(ce' + c'e) - 20dd',$$

which has an obvious connection with

$$ag - 6bf + 15ce - 10d^2;$$

and the same is the case for functions of any order.

The following theorem is easily verified; but I am unacquainted with the general theory to which it belongs.

“If U, V are any functions of the second order, and $W = \lambda U + \mu V$; then

$$B'_2[B_2(W, W), B_2(W, W)] = 0$$

(where B'_2 relates to λ, μ) is the same that would be obtained by the elimination of x, y between $U = 0, V = 0$.”³ (See Note.)

In fact this becomes

$$4(ac - b^2)(a'c' - b'^2) - (ac' + a'c - 2bb')^2 = 0,$$

which is one of the forms under which the result of the elimination of the variables from two quadratic equations may be written. This is a result for which I am indebted to Mr Boole.

Passing to the third degree, we may consider in particular the derivatives

$$\square UVW = 23^\alpha 31^\alpha 12^\alpha UVW = C_\alpha(U, V, W);$$

writing for shortness

$$\delta_{x_2}^{\alpha-t} \delta_{y_2}^t \delta_{x_3}^{\alpha-r} \delta_{y_3}^r U = U^{-r,t},$$

we have the general term

$$\pm U^{-r,t} V^{-s,r} W^{-t,s},$$

where r, s, t extend from 0 to α . By changing the suffixes r, s the following more convenient formula

$$\pm U^{-\rho,t} V^{-\sigma,\rho} W^{-t,\sigma},$$

where t extends from 0 to 2α : ρ, σ , and $3\alpha - \rho - \sigma$ must be positive and not greater than 2α .

In particular, according as α is odd or even,

$$C_\alpha(U, U, U) = 0,$$

³Not given with the present paper.

$$C_{\alpha}(U, U, U) = 6\Sigma\Sigma\{(-)^{t+s} U^{-\rho,t} U^{-\sigma,\rho} U^{-t,\sigma}\} - \frac{1}{2}\left[(-)^{\frac{\alpha}{2}} U^{\frac{\alpha}{2},\frac{\alpha}{2}}\right]^3,$$

omitting therein those values of ρ, σ for which $\rho > \sigma$ or $\sigma > 3\alpha - \rho - \sigma$, and dividing by two the terms in which $\rho = \sigma$ or $\sigma = 3\alpha - \rho - \sigma$, and by six the term for which

$$\rho = \sigma = 3\alpha - \rho - \sigma, = \alpha.$$

In particular, for functions of the fourth or eighth orders we have the constant derivatives

$$ace - ad^2 - b^2e - c^3 + 2bcd;$$

$$aei - 8bd - 4afh + 8ag^2 + 8ic^2 + 12beh - 8chd - 8bgf - 22ceg + 24cf^2 + 24d^2g - 36def + 15e^3$$

the first of which is a simple determinant. Thus we have been led to the functions $ae - 4bd + 3c^2$ and $ace - ad^2 - eb^2 - c^3 + 2bcd$, which occur in my “Note sur quelques formules &c.” (*Crelle*, vol. *xxix*. [1845] [15]), and in the forms which M. Eisenstein has given for the solutions of equations of the first four degrees.

Let U be a function of the order 4α : the derivative C may be expressed by means of the derivatives B . For, consider the function

$$B_a[U, B_a(V, W)];$$

paying attention to the signification of B , this may be written

$$10^{-2a} 23^{2a} UVW,$$

where the symbols ξ_0, η_0 refer to the two systems x_2, y_2 : x_3, y_3 . Thus it is easily seen that we may write

$$\xi_0 = \xi_2 + \xi_3, \quad \eta_0 = \eta_2 + \eta_3, \text{ or } 10 = 12 + 13 = 12 - 31,$$

whence the function becomes

$$(12 - 31)^{2a} 23^{2a} UVW,$$

of which all the terms vanish except

$$12^a 23^a 31^a UVW.$$

$$[4a]^2 \div [2a]^2 \dots [a]^a$$

Hence putting

$$K = \frac{[2a]^{2a}}{[2a]^a} = \frac{2^{2a} \cdot 1 \cdot 3 \dots (4a - 1)}{2 \cdot 4 \dots 4a},$$

we have

$$B_a [U, B_a(V, W)] = K C_a (U, V, W),$$

or in particular

$$B_a [U, B_a(U, U)] = K C_a (U, U, U).$$

Thus for example, neglecting a numerical factor,

$$\begin{aligned} & (ax^2 + 2bxy + cy^2)(cx^2 + 2dxy + ey^2) - (bx^2 + 2cxy + dy^2)^2 \\ &= (ac - b^2)x^4 + 2(ad - bc)x^3y + (ae + 2bd - 3c^2)x^2y^2 + 2(be - cd)xy^3 + (ce - d^2)y^4, \end{aligned}$$

and then

$$\begin{aligned} & e(ac - b^2) - 4d(ad - bc) + 6c(ae + 2bd - 3c^2) - 4b(be - cd) + a(ce - d^2) \\ &= 3(ace - ad^2 - b^2e - c^3 + 2bcd). \end{aligned}$$

We have likewise the singular equation

$$B_{2a}(V, W) = K \left(\frac{d^a}{dx^a} \frac{d^a}{dy^a} - \dots \pm \frac{d^{2a}}{dx^{2a}} \right) c_a(U, V, W)$$

where

$$U = \frac{1}{[4a]} \left(ax^{4a} - \frac{[4a]}{[1]} bx^{4a-1}y \dots + zy^{4a} \right) \cdot \&c.$$

If however $U = V = W$, we must write

$$B_{2a}(U, U) = \frac{1}{6} K \left(\frac{d^{2a}}{dx^{2a}} \frac{d^{2a}}{dy^{2a}} - \dots \pm \frac{d^{4a}}{dx^{4a}} \right) c_a(U, U, U),$$

the reason of which is easily seen. This subject will be resumed in the sequel.

The functions C may be transformed in the same way as the functions B have been. In fact

$$C_a(U, V, W) = 12^{a-k} 23^{a-k} 31^{a-k} C_k(U, V, W);$$

if in particular $k = 1$, then

$$\dots U^{a-k} \text{ for } U^{a-k}, \text{ \&c.}$$

but in general

$$\xi_1^\rho \eta_1^\sigma \xi_2^{\rho'} \eta_2^{\sigma'} \xi_3^t \eta_3^{t'} C_k(U, V, W), \text{ where } \rho + \rho' = \sigma + \sigma' = t + t' = 2a - 2,$$

$$2a - 2, \quad 4a - 4, \quad 2a - 2, \quad U^{\cdot\rho} V^{\cdot\sigma} W^{\cdot t} \text{ for } U^{\cdot\rho}, \text{ \&c.}$$

whence

$$C_a(U, V, W) = \Sigma \Sigma \{ (-)^{r+t} U^{\cdot\rho} V^{\cdot\rho+1} V^{\cdot\sigma} W^{\cdot 3a-\rho-\sigma-1} \} \cdot \Sigma [(-)^t A_r^t A_{\sigma+1-r} A_t],$$

where $A_p^t = \frac{[a-1]^p}{[t]^p}$; t extends from 0 to $a-1$; $\rho, \sigma-1$, and $3a-\rho-\sigma-2$ may have each of them any positive values not greater than $2a-2$.

In particular

$$C_a(U, U, U) = 6 \Sigma \Sigma \{ (-)^{r+t} U^{\cdot\rho} U^{\cdot\sigma-1} U^{\cdot 3a-\rho-\sigma-2} \} \cdot \Sigma [(-)^{r+t} A_r^t A_{\sigma+1-r} A_t],$$

where ρ, σ need only have such values that $\rho < \sigma-1, \sigma-1 < 3a-\rho-\sigma-2$.

In particular the derivative $aei - \dots + 15e^3$ may be transformed into

$$\begin{vmatrix} a, & d, & g \\ b, & e, & h \\ c, & f, & i \end{vmatrix} - \begin{vmatrix} a, & e, & f \\ b, & f, & g \\ c, & g, & h \end{vmatrix} - \begin{vmatrix} b, & c, & g \\ c, & d, & h \\ d, & e, & i \end{vmatrix} + 6 \begin{vmatrix} b, & d, & f \\ c, & e, & g \\ d, & f, & h \end{vmatrix} - 15 \begin{vmatrix} c, & d, & e \\ d, & e, & f \\ e, & f, & g \end{vmatrix}$$

in which form it is obviously a linear function of the determinants